From Pixels to Agents: Calibrating a Heterogeneous NetLogo Traffic Model from YOLOv8 Vehicle Tracks

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Abstract—Microscopic traffic simulation is widely used to evaluate signal-control strategies at urban intersections, but its usefulness depends on realistic demand and vehicle-behavior inputs. Simulation inputs are often derived from aggregate counts and default parameters, while differences among vehicle classes may be inadequately represented. In addition, errors introduced by computer-vision systems are not always examined in terms of their effects on simulation outputs. This proposed study will develop an offline pipeline to extract class-labelled vehicle trajectories from fixed-camera video using YOLOv8-based detection and multi-object tracking. Image coordinates will be mapped to a road-plane coordinate system through homography, allowing the extraction of per-lane arrival, queue, speed, and discharge-headway features. These features will inform the calibration of a heterogeneous agent-based traffic model in NetLogo, in which cars, two-wheelers, and heavy vehicles will be represented using class-specific Intelligent Driver Model parameters. The proposed evaluation will compare class-aware calibration with homogeneous and class-mix-only baselines using held-out observations. It will also examine fixed-time, vehicle-actuated, and queue-adaptive signal-control strategies, and investigate how detection misses, class-confusion errors, and count bias may affect simulated outcomes. The study aims to provide a reproducible framework for examining the effects of vehicle-class heterogeneity and vision uncertainty on intersection-control evaluation.

Index Terms—agent-based modeling, traffic simulation, model calibration, YOLOv8, multi-object tracking, heterogeneous traffic, adaptive signal control, perception error

I. INTRODUCTION

Signalized intersections are among the most consequential locations in urban road networks because small changes in signal timing, queue discharge, and vehicle composition can produce substantial changes in delay, throughput, and queue spillback. Microscopic traffic simulation is therefore commonly used to examine intersection-control policies before field deployment. However, the credibility of these simulations depends strongly on the quality of their inputs, including observed demand, signal timing, road geometry, and the behavioral rules that determine how vehicles accelerate, follow one another, and discharge from a queue.

A common limitation in simulation-based intersection studies is the use of aggregate traffic counts and default driver-behavior parameters. Such models may treat all vehicles as equivalent passenger cars, even where two-wheelers, buses, and trucks constitute a substantial share of traffic. This simplification can be problematic at signalized approaches, where vehicle length, acceleration capability, desired time headway, and minimum spacing influence queue discharge and saturation flow. A homogeneous model may reproduce total throughput while misrepresenting the timing and composition of queue release, potentially biasing estimates of delay and the relative performance of signal-control strategies.

Camera-based perception has made it increasingly feasible to obtain vehicle-level trajectories from ordinary traffic video. Object detectors identify vehicles in individual frames, while multi-object trackers associate detections across frames to form trajectories [1], [2]. When camera geometry is calibrated, these trajectories can be transformed into ground-plane positions and used to estimate flows, speeds, headways, queue lengths, and discharge patterns. This creates an opportunity to inform microscopic traffic simulation using observed, class-labelled trajectories rather than relying exclusively on manual counts or generic parameter values.

However, vision-derived data are imperfect. Detectors may miss vehicles, especially small or distant objects; trackers may fragment trajectories or switch identities; and vehicle classes may be confused, particularly in mixed-traffic environments. If these errors are passed into a traffic model without examination, they may distort demand estimates, queue-length measurements, and the comparison of signal-control alternatives. Detection and tracking performance therefore needs to be considered alongside the effects that perception errors may have on downstream simulation outcomes.

To investigate these issues, the proposed study will consider an integrated offline pipeline. Fixed-camera video from a signalized approach will be processed using YOLOv8-based detection and ByteTrack-based multi-object tracking [8], [2]. Detected vehicles will be grouped into three classes—cars, two-wheelers, and heavy vehicles—and their trajectories will be mapped from image coordinates to metric road-plane coordinates using a homography estimated from measured ground-control points. The resulting trajectories will be used to derive per-lane and per-class arrival headways, speeds, queue lengths, discharge headways, and class shares. These features will inform a heterogeneous agent-based model in NetLogo, where vehicle agents will follow class-specific Intelligent Driver Model (IDM) parameters and interact with a signal agent.

The proposed study is guided by three research questions. First, can class-aware calibration from video reduce error against held-out observations of queue length, discharge headway, and speed distribution compared with a homogeneous model calibrated on aggregate flow? Second, how sensitive might simulated delay and queue outcomes be to perception errors, including missed vehicles, class misclassification, and count bias? Third, can queue-adaptive signal-control strategies reduce delay relative to fixed-time control under calibrated demand, and would any observed advantage remain under perception error?

The intended contributions are threefold. First, the study will develop a reproducible pipeline connecting monocular traffic video, class-labelled trajectory extraction, ground-plane feature extraction, and a heterogeneous NetLogo agent-based model. Second, it will compare homogeneous, class-mix-only, and class-aware calibration using held-out validation data. Third, it will examine how detection misses, class confusion, and count bias may propagate into simulated delay and the relative ranking of signal-control strategies. These are proposed contributions; their extent will depend on the results obtained during implementation and evaluation.

The proposed work is scoped to one signalized approach, one camera viewpoint, and three vehicle classes. It will use offline processing rather than real-time inference, and it will not claim city-wide generalization. The remainder of this paper reviews related work and establishes the background for the proposed vision-based calibration and simulation study.

II. LITERATURE REVIEW

This section reviews prior work relevant to the proposed study: vehicle detection and multi-object tracking; camera geometry and traffic-feature extraction; microscopic traffic and agent-based modeling; and simulation calibration, signal control, and perception-error propagation. It concludes by identifying the specific intersection of these research areas that motivates the proposed work.

A. Vehicle Detection and Multi-Object Tracking

Video-based traffic analysis typically begins with object detection, followed by data association across frames. The You Only Look Once (YOLO) family of detectors formulates object detection as a single-network prediction problem, enabling relatively fast inference while retaining competitive accuracy. The original YOLO paper established this unified detection approach and reported real-time performance on standard benchmarks [1]. YOLOv8 is a later detector implementation in this family and is documented by Ultralytics [8]. Its suitability for a particular traffic scene should be assessed using the camera viewpoint, vehicle mix, and image conditions of the target site.

Detection alone does not produce trajectories. Multi-object tracking associates detections over time, assigning consistent identities to the same physical vehicle. ByteTrack associates both high- and low-confidence detection boxes rather than discarding low-confidence detections outright; this design is intended to preserve objects through occlusion, motion blur, and detection fluctuations [2]. In traffic scenes, however, occlusion between vehicles, small object size at a distance, and class confusion remain potential sources of error. A traffic-analysis pipeline should therefore evaluate class-specific counting performance against manually annotated ground truth rather than relying only on generic detection metrics.

Annotated video benchmarks can support the evaluation of detection and tracking methods. However, benchmark footage may differ from a particular signalized approach in camera geometry, local vehicle composition, and the visibility needed for physically meaningful speed and queue estimation. Benchmark performance should therefore be interpreted in relation to the target scene, and site-specific validation remains important when the intended application involves traffic measurements.

B. From Image Coordinates to Traffic Features

A detector and tracker produce image-space bounding boxes and track identities, but traffic-simulation calibration requires physically meaningful quantities such as speed in meters per second, queue length in vehicles, and headway in seconds. Camera calibration is therefore essential. A homography maps points on a planar road surface from image coordinates to ground coordinates, provided that the road is approximately planar and that reference points with known ground positions are available.

For the proposed study, ground-control points will be measured at the site, including lane boundaries, stop-line endpoints, and road-marking features. The homography will be estimated from these correspondences and evaluated using held-out reference points. Vehicle positions will be represented by the bottom-center of each detection bounding box, which lies closer to the road plane than the geometric box center. After mapping, trajectories will be filtered and smoothed before speed, headway, queue-length, and discharge-headway features are computed.

This feature-extraction stage is important because calibration targets must be operationally defined. In the proposed study, queue length is intended to represent the number of vehicles travelling below a specified speed threshold within a defined distance upstream of the stop line. Discharge headway is intended to be measured between consecutive vehicles crossing the stop line after green onset. Defining these quantities in advance will help make the simulated and observed measures comparable and reduce ambiguity during evaluation.

C. Microscopic Traffic and Agent-Based Modeling

Microscopic traffic models represent individual vehicles and their interactions, making them appropriate for studying queue formation, discharge, and signal-control effects. Car-following models describe longitudinal motion based on a vehicle's speed, the gap to its leader, and their relative speed. The Intelligent Driver Model (IDM) is a widely used continuous car-following model that combines desired speed, desired time headway, minimum gap, maximum acceleration, and comfortable deceleration in an acceleration rule [3]. Its parameters have behavioural interpretations, making the model a potential basis for calibration using trajectory-derived observations.

Agent-based modeling provides a complementary way to represent traffic. Rather than describing traffic solely through aggregate flow relationships, an agent-based model represents each vehicle as an agent with state variables and behavioral rules. NetLogo supports spatial agents, stochastic processes, and simulation experiments [7]. These features make it a possible environment for representing vehicle agents, a signal agent, demand generation, and control logic in an explicit and modifiable form.

The proposed model will represent vehicle heterogeneity through three classes: cars, two-wheelers, and heavy vehicles. These classes differ in physical length and may differ in desired speed, acceleration, time headway, and minimum-gap parameters. Lane-changing behavior is outside the initial scope to keep the model focused on longitudinal interactions and to limit the number of parameters requiring calibration. The MOBIL framework provides a general approach to extending car-following models with lane-changing rules, should this behavior be considered in later work [4].

D. Calibration and Validation of Traffic Simulation

Calibration is the process of selecting model parameters so that simulated outputs match observed system behavior. Hollander and Liu emphasize that microsimulation calibration requires attention to the scope of calibration, parameter selection, objective formulation, goodness-of-fit measurement, and repeated stochastic runs [5]. These principles are relevant to the proposed study, in which the calibration targets and parameter search would need to be defined before model fitting, with repeated runs used to account for stochastic variation.

A central methodological requirement is the separation of calibration and validation data. Parameters should be fitted using a calibration dataset, while model quality should be assessed on separate held-out observations. This separation reduces the risk that a model merely reproduces patterns specific to the data used for fitting. The proposed study will consider three model variants: a homogeneous baseline with one shared parameter set, a class-mix-only model that includes observed vehicle shares and lengths but shared behavior, and a class-aware model with class-specific IDM parameters. This comparison is intended to distinguish the effect of representing vehicle composition from the effect of representing class-specific driving behavior.

Potential calibration targets include class-specific speed distributions, discharge headways, maximum queue length, and per-cycle throughput. A normalized objective function could allow quantities with different units and scales to contribute comparably. Parameter uncertainty should also be considered; retaining several near-optimal parameter sets may be more informative than presenting a single configuration as uniquely identifiable.

E. Signal Control and Simulation-Based Evaluation

Signal-control evaluation requires a meaningful baseline and a control strategy whose performance can be compared under consistent demand and stochastic conditions. Fixed-time control provides a standard baseline because its cycle length and green splits can be observed or optimized for a given demand level. Vehicle-actuated control extends green phases in response to detected vehicles, while queue-based adaptive control allocates green time according to estimated queue conditions.

Max-pressure control provides a theoretical foundation for adaptive signal control by selecting phases that reduce network pressure, defined through upstream and downstream queue conditions [6]. The proposed study does not require a full max-pressure controller; rather, its queue-adaptive strategy is intended to draw on the general principle of allocating service according to queue conditions. The planned comparison may include an observed fixed-time plan, an optimized fixed-time plan, vehicle-actuated control, and queue-based adaptive control. The final set of strategies will depend on the scope that can be implemented and evaluated reliably.

A limitation of using a single-camera view of one approach is that conflicting demand from other approaches may not be directly observable. One possible modeling assumption is to represent the conflicting phase as a deterministic point queue with scenario-based arrival rates. If this assumption is used, it should be stated explicitly and treated as a limitation; control conclusions would then be conditional on the assumed conflicting demand.

F. Perception Error Propagation

An important issue for the proposed study is how perception error may affect traffic-simulation outcomes. Detection and tracking studies commonly report measures such as precision, recall, multiple-object-tracking accuracy, identity F1 score, and higher-order tracking accuracy. These measures are useful for evaluating a vision module, but they do not directly indicate how a missed motorcycle, a misclassified truck, or a biased vehicle count might affect simulated delay or the comparison between signal-timing policies.

The proposed study will examine this issue by considering controlled perturbations to trajectory-derived features rather than modifying the simulation rules themselves. Missed vehicles could be represented by randomly removing a proportion of tracks; class confusion could be represented using a measured or scaled confusion matrix; and count bias could be introduced by scaling arrival rates. Applying the same perturbations across the model variants and control strategies would help examine whether perception errors alter simulated outcomes or the relative ranking of the strategies.

G. Research Gap

Prior research has developed methods for vehicle detection and tracking, microscopic car-following models, simulation calibration, and adaptive signal control. The proposed study focuses on the intersection of these areas: class-labelled trajectory extraction from fixed-camera video, ground-plane feature estimation, class-specific agent-based behavior, held-out validation, and examination of how perception errors may affect signal-control evaluation. The contribution is therefore framed around an integrated

evaluation approach rather than novelty in any individual component.

The reviewed areas also point to a practical distinction between evaluating a vision system and evaluating the traffic conclusions drawn from its output. Detection, tracking, and counting metrics describe perception performance, while simulation studies may rely on manually collected or assumed demand and behavior parameters. A class-aware calibration approach offers a way to examine whether vehicle composition and class-specific behavior matter for queue discharge. Testing perception errors within the same evaluation framework may further show whether apparent differences between control strategies are sensitive to imperfect inputs. These questions motivate the proposed study, but their answers will depend on the data and experiments ultimately completed.

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